

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of Bachelor of Technology (B.Tech) (C E / IT)

University Examination November 2010

MA201 Discrete Mathematics

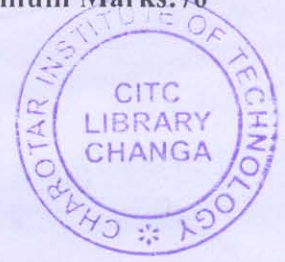
Date: 25.11.10, Thursday

Time: 10:30 a.m to 01:30 p.m

Maximum Marks: 70

Instructions:

1. The question paper contains two sections. Each section has three questions.
2. Each section carries 35 marks.
3. Figures to the right indicate marks.
4. Make suitable assumption whenever necessary and state them.



SECTION- I

FOR REFERENCE

Q.1 Do as Directed.

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1. Give an example of a relation which is symmetric as well as antisymmetric.
2. Define: Sub boolean algebra
3. What rules of inference are used in this famous argument? "All men are mortal. Socrates is a man. Therefore, Socrates is mortal."
4. Define: Anti-atoms.
5. Let $P(n)$ is the proposition "If n is a positive integer greater than 1, then $n^2 > n$." To prove $P(0)$ is true, what kind of proof did you use?
6. Every distributive lattice is a Boolean algebra. True or False.
7. Let $P(X)$ be the collection of all subsets of a non empty set X with at least three elements. Let $A, B \in P(X)$. If we define relation R such that A and B are in the relation if $A \cap B = \emptyset$, then R is not antisymmetric. True or False.

Q.2 (a) Write distributive laws and De Morgan's laws for the logical equivalences. By using these show that $\neg(p \vee (\neg p \wedge q))$ and $\neg p \wedge \neg q$ are logically equivalent. 3

(b) 1. Explain: Hasse diagram and draw the Hasse diagram of $(S_{100}, /)$. 3

2. Prove that Every chain is a distributive lattice. 2

(c) 1. Define: atoms and antiatoms. Find the join irreducible element and meet irreducible element of the following: 3

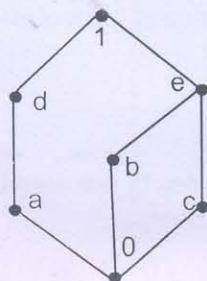
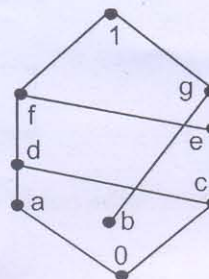


figure 1



2. Define: Boolean algebra. Prove that $(S_{30}, *, \oplus, ', 0, 1)$ is a Boolean algebra. 3

OR

Q.2 (a) Define: Tautology and Contradiction. Check whether the following are tautologies or not: $(\neg p \wedge (p \rightarrow q)) \rightarrow \neg q$ and $(\neg p \wedge (p \rightarrow q)) \rightarrow \neg p$. 3

(b) 1. Define: Poset. Prove that $(P(A), \subseteq), A \neq \emptyset$ is a poset. 3

2. Explain the duality of a lattice. 2

(c) 1. Define: the join irreducible element and meet irreducible element. Find atoms and antiatoms of the posets given in the figure 1: 3

2. Explain minterm and maxterm of a Boolean expression. 3

Q.3 (a) Let $F(x,y)$ be the statement “x can fool y.” where the universe of discourse is the set of all people in the world. Use quantifiers to express each of the following statements: 3

- (i) There is no one who can fool everybody.
- (ii) Everyone can be fooled by somebody.
- (iii) There is exactly one person whom everybody can fool.
- (iv) Everyone can fool everybody.

- (b)** 1. Prove that the direct product of two lattices is also a lattice. 3
 2. Define: Minimal and Maximal elements of a poset. Find the minimal and maximal element of the poset $(\{2, 4, 6, 12, 20\}, /)$. 2

- (c)** 1. Define: equal Boolean expressions. Let $(B, *, \oplus, ', 0, 1)$ be a Boolean algebra. 3
 Then prove that $(a' * b' * c) \oplus (a * b' * c) \oplus (a * b * c') = b' * (a \oplus c)$
 2. find all the minterm and maxterm of a Boolean algebra with three variables. 3

OR

- (a)** Let $P(x,y)$ be the statement “x has taken class y.” where the universe of discourse for x is the set of all students in your class and for y is the set of all computer science courses at your school. Express each of the following quantifications in English: 3

- (i) $\exists x \exists y p(x, y)$, (ii) $\exists x \forall y p(x, y)$, (iii) $\forall x \exists y p(x, y)$, (iv) $\exists y \forall x p(x, y)$,
 (v) $\forall y \exists x p(x, y)$, (vi) $\forall x \forall y p(x, y)$

- (b)** 1. Define: the direct product of two lattices and draw the Hasse diagram of $(S \times S \times S, \leq)$, $S = \{0, 1\}$. 3
 2. Define: GLB and LUB of a set in the poset. Find all possible upper bounds and lower bounds of the set $\{\{1, 2\}, \{1, 4\}, \{3, 4\}\}$, a subset of the poset $(\{\{1\}, \{2\}, \{4\}, \{1, 2\}, \{1, 4\}, \{2, 4\}, \{3, 4\}, \{1, 3, 4\}, \{2, 3, 4\}\}, \subseteq)$. 2

- (c)** 1. Draw the switching circuit of $(p + q) \cdot (p + t) + t \cdot (p + (q \cdot t))$. Find the equivalent simplified Boolean expression and draw the simplified circuit. 3
 2. Find a minimal sum of product form of the Boolean expression $xyz + xyz' + xy'z + x'yz + x'yz'$. Verify your answer by using Karnaugh map. 3

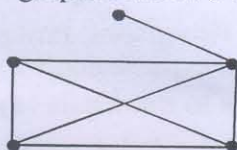
SECTION – II

Q. 4 Do as directed:

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1. Define: Binary tree
2. Isomorphic graphs may not preserve the incidence relation. True or False.
3. How many vertices are there in a tree with 19 edges?
4. Construct a graph G with 6 vertices such that G has exactly 5 edges but G is not a tree.
5. State Euler's theorem.
6. In the adjacency matrix of a graph, the number of 1's in each column equals the degree of the corresponding vertex. True or False.
7. Define: Hamilton graph.

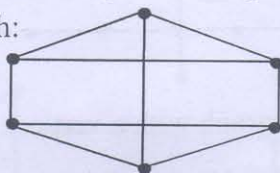
- Q. 5 (a)**
1. Prove that the sum of the degrees of all vertices in G is twice the number of edges in G . 2
 2. Is it possible to draw a simple graph with 7 vertices of degrees 1, 3, 3, 4, 5, 6, 6? Justify your answer. State the result which you used here. 3
- (b)**
1. Define a cut set of a graph. Find two cut sets of the following graph: 3



2. Define: Complete graph. Prove that a complete graph K_n has an Eulerian circuit if n is odd. 3
- (c)** Explain stereographic projection. 3

OR

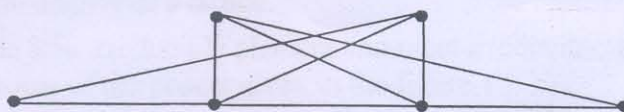
- Q. 5 (a)**
1. Prove that the number of vertices of odd degree in a graph is always even. 2
 2. How many vertex do the graph has if it contain 21 edges, 3 vertices of degree 4 and others each of degree 3? State the result which you used here. 3
- (b)**
1. Define: Isomorphic graphs. Is the graph of $K_{3,3}$ isomorphic to the following graph: 3



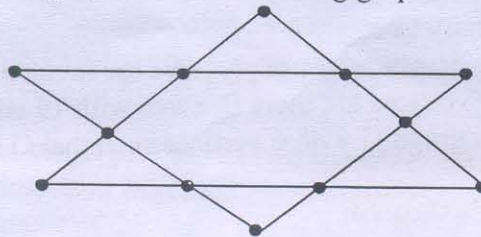
2. Draw the graph represented by the given adjacency matrices: 3

$$(i) \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}, (ii) \begin{pmatrix} 0 & 1 & 3 & 0 & 4 \\ 1 & 2 & 1 & 3 & 0 \\ 3 & 1 & 1 & 0 & 1 \\ 0 & 3 & 0 & 0 & 2 \\ 4 & 0 & 1 & 2 & 3 \end{pmatrix}$$

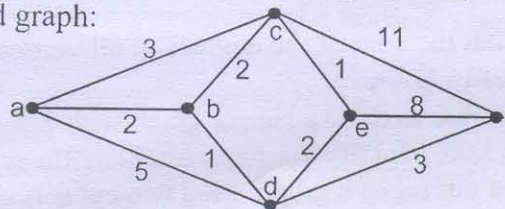
- (c)** Define: Planar graph. Draw the planar representation of the following graph: 3



- Q. 6 (a)**
1. Define: Euler graph. Give an example of a Hamilton graph which is not an Euler graph. Justify your answer. 2
 2. Define: Subgraph. Draw two edge disjoint subgraphs and two vertex disjoint subgraphs of the following graph: 3



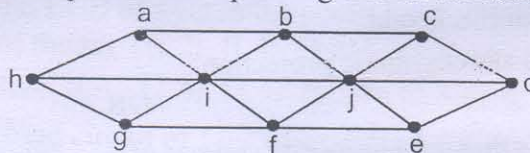
- (b)**
1. Define: Connected graph. Find the connected component of the graph $G=(V,E)$, $V=\{1,2,3,4,5,6\}$ and $E=\{(1,5),(2,3),(6,5),(4,6)\}$. 3
 2. Find the minimal spanning tree by prim's algorithm for the following weighted graph: 3



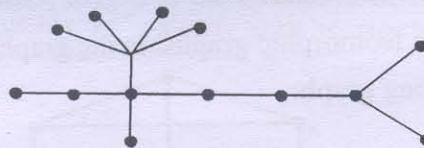
- (c)** Draw Kuratowski's two graphs. Prove that the complete graph of five vertices is non-planar. 3

OR

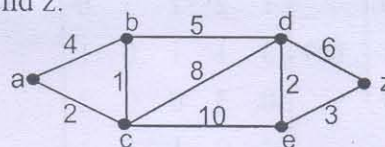
- Q. 6 (a)**
1. Define: Spanning tree. Draw one spanning tree and find the fundamental cut set with respect to that spanning tree of the following graph: 2



2. Define: Eccentricity of a vertex in a graph. Find eccentricity of each vertex and center of the following graph: 3



- (b)**
1. Prove that a simple graph with n vertices and k components can have at most $\frac{(n-k)(n-k+1)}{2}$ edges. 3
 2. Use dijkstra's algorithm to find the length of a shortest path between the vertices a and z : 3



- (c)** Prove that in any simple, connected planar graph with f regions, n vertices and e edges, $e \leq 3n - 6$. And hence prove that K_5 is non-planar. 3